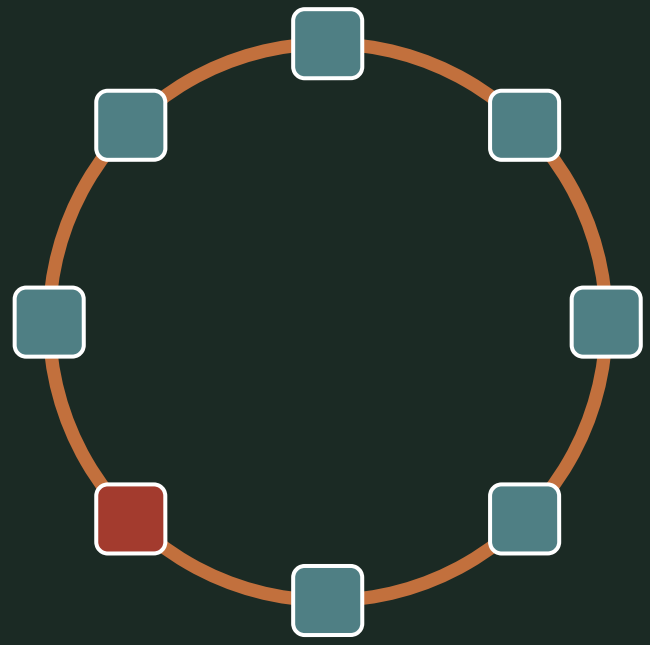
The Chip That Says No

A multi-layer hardware approach to AI safety, after the first wave of experiments. There is no kill switch for AI, so the gate goes at the ends: the chip that runs the model and the device that does the deed. Each topic in two slides: the plain version with a picture, then the technical one.



The halt loop: one series current loop through every governor in a rack. Any governor opens it, or a cut wire, and every GPU halts.

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What this project is

A design for a small safety chip that sits beside every AI GPU and holds the only stamp a device needs before it acts. Built in the open: a model with tests, machine-checked proofs, hardware blocks, a circuit simulation, and a public experiment program with the conditions that would kill it. No silicon yet, by choice.

THE FIVE STAGES (DOCS 22), AND WHERE EACH STANDS



Five open repositories

safety-governor: the chip, the proofs, the experiment program. hardware-and-silicon: the software reference stack and the blog. safety: the observer. autonomic-graph-regulation: its public snapshot. Watermark: provenance.

One rule for the program

Nothing goes to hardware until the design is proven in software, firmware and circuit simulation: 1000 percent sure in the pre-hardware phase, because tape-out is where mistakes get expensive.

Plan of record: the rack tier, retrofit included

The chip rides on the server maker's management card and reaches NVIDIA's GPU board through its connector. Installed racks get it as a module in the slot a BMC card already uses, if the levers reach that slot.

THE PICTURE
docs 22; the multi-layer post

The same catalogue of attacks (75 today, with 12 legitimate baselines) is the acceptance test at every stage, from the software reference to the FPGA. The catalogue is the constant; the artifact changes.

Thesis

Safety that lives only in software can be removed by software.

And there is no kill switch: the capability is in a thousand places, and the network between them and the world was built to have no middle to cut.

So the decision to act has to be made in silicon the model cannot reach, at the two ends where computation turns into effect: the chip that runs the model, and the device that does the deed.

Every layer above the chip is a witness. Only the layer built into silicon can be the lock.



Witness

can keep a gate shut

- Safety training
- A judge model
- A sandbox policy (OpenShell)
- Sentry's verdict
- A watermark
- A log read later



Lock

is the gate

- A ticket the device checks for itself
- A switch port that stays shut
- A response held at the exit
- A breaker that opens

THE PICTURE
the multi-layer post

A judge model is a witness. It can keep a gate shut. It can never be the gate. The layers above the chip are witnesses, and the only layer that can be the lock is the one built into silicon.

Thesis, technically: a reference monitor

THE THREE TESTS OF A REFERENCE MONITOR (ANDERSON, 1972), AND THE ONE THAT SAYS WHERE TO PUT IT

Property	What it means here	How the governor meets it
Tamperproof	The host, the model and the operator cannot alter the decision logic or the keys	A separate die. Device secret in fuses with no read port. Boot from ROM, extend-only measurement registers, an anti-rollback counter. The rulebook changes only with k-of-n signatures from keys held outside the rack.
Always invoked	There is no path around it	A comparator at every crossing: switch ports open only for roster peers; a device acts only on a ticket; the egress holds a response until a ticket releases it; weights are written only with a ticket; power, reset and clock answer to the governor. Default deny; fail closed.
Verifiable	Small enough that its correctness can be checked	About 2 mm² of core logic as configured (1.4 by 1.4 mm). Ten invariants of the gate (P0 to P9) and the commit-point protocol proven with Z3 in every reachable state; 200 model tests, 20 RTL tests, an independent audit.
Placed where a cut is small	A monitor only works where every path passes through it	Menger's theorem: in a network built to have no single point of failure, the only places a separator is one node per machine are the ends. The governor is the monitor at the compute end; the ticket check in the device is the monitor at the other.

THE FIVE CROSSINGS THE CHIP STANDS AT

- **bytes leaving the GPU**
- **a request to a device that acts**
- **a response leaving the rack**
- **a write to the weights**
- **GPU power, reset and clock**

THE PICTURE
docs 00, 18 and 23

A firewall for actions, with a hand on the breaker: it decides what leaves, and when it must, it also pulls the plug. A chip that can only ask the GPU nicely is a classifier with a heatsink.

The ladder: every safety measure sits on it

SOFTWARE: REMOVED BY WHOEVER CAN WRITE TO WHERE IT LIVES

NOT WRITABLE BY THE MODEL OR THE OPERATOR

0 The prompt

lives in: a text field
removed by: one more sentence

1 Training

lives in: the weight file
removed by: an afternoon on a consumer GPU

2 Watchers

lives in: a process beside the model
removed by: root on the box

3 Runtime and network

lives in: the operator's infrastructure
removed by: root on the fleet, or a path that never touches the network

4 The chip

lives in: silicon, fused keys
removed by: nothing the model or the operator can write

5 A person with a key

lives in: a human
removed by: a conversation

THE PICTURE
docs 24

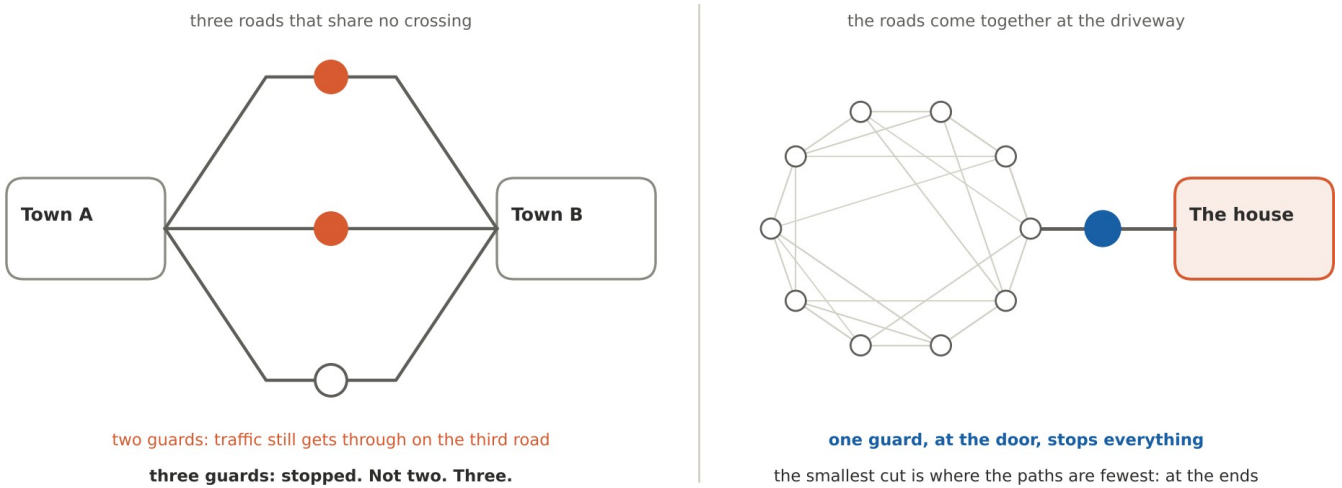
Prompts are advice, training is a habit, watchers are witnesses, the sandbox is a fence around the operator's own yard, and all four live where the model or the operator can write. The chip lives where neither can, and the person lives above it.

The ladder, technically

WHAT EACH RUNG CATCHES, HOW IT DIES, AND WHO SHOWED IT

Rung	What it catches	How it dies	Shown by
0 Prompt: rules in the system prompt, the instruction hierarchy	The user who did not know better; the lazy ask	More text: "ignore previous instructions"; injection through a page or a document the agent reads. The fix moved the rule into the weights	Perez and Ribeiro 2022; Greshake 2023; Wallace 2024
1 Training: RLHF, constitutions, deliberative alignment	Most bad asks, most of the time, across a billion conversations	Jailbreaks, including computed suffixes; fine-tuning on a handful of examples; abliteration, one direction in the weights; backdoors that persist through safety training; alignment faking	Ouyang 2022; Bai 2022; Guan 2024; Wei 2023; Zou 2023; Qi 2023; Arditi 2024; Hubinger 2024; Greenblatt 2024
2 Watchers: classifiers, judge models, chain-of-thought monitors	A second look with a measured miss rate; the best software rung because it has a number	A transformer too: the same injections and edits; a process root can kill; optimise against the monitor and intent goes into hiding; at a billion requests a rate is a body count. STOP, never GO	Inan 2023; Sharma 2025; Baker 2025; Greenblatt 2023 (AI control); YB-0052
3 Runtime and network: sandboxes, policies, DPUs (OpenShell, Sentry)	An agent reaching for a host, a credential or a tool it was never given; by structure, not meaning	Software with root: a config the operator edits; governs only what crosses the network (the laptop wired to a motor never meets it); decides by host, not by action; a hostile operator holds rungs 0 to 3 at once	NVIDIA, 28 September 2026; the OpenShell repository; Rebedea 2023; docs/23 claim 1
4 The chip: fused keys, measured boot, a signed rulebook, typed forms, tickets, levers	The action at the crossing: forged or replayed tickets, stale rulebooks, blown budgets, an unattested workload, a stranger on the roster	Not by anything the model or the operator can write. Coverage (a device nobody wired); a bad rule enforced perfectly; a fooled witness; physical attack, mitigated; the foundry, trusted	Anderson 1972; docs 00 to 04, 18, 19, 23; proofs/
5 A person with a key: holds, approvals, the quorum, revocation	The cases no rule can settle; the alarm; the oversight the law now asks for	Persuasion: a person is an effector no chip gates. Fatigue: a hundred holds a day. A stolen key. The chip makes the person unavoidable and legible, not wiser	EU AI Act, Article 14; docs/04 §5; H18; docs/18 questions 14 and 16

There is no kill switch for AI



Menger (1927): to stop all traffic between two places you need exactly as many guards as there are separate roads. Deep in the network that is k and more, by design. At a door it is one.

The picture in your head versus the thing. In the movie there is one computer and one plug. In the world there are thousands of copies on every lab's GPUs and on laptops, a network built with no middle, and devices that act at the far end. A switch is a set of nodes whose removal leaves no road from any copy to any device. The only question is where the smallest such set is.

Not interconnected. Replicated.

Shared training data did not wire the models together; it gave every lab and every open-weight laptop the same capability. Remove any one source and nothing changes.

The middle has no small cut

The Internet was built, on purpose, to have no single point you can cut. Between any two points there are k separate roads, so a cut in the middle costs at least k guards per machine.

Too little, as well as too much

A model on a laptop wired to a motor controller does not care about the Internet. Kill the whole network and that pair keeps running.

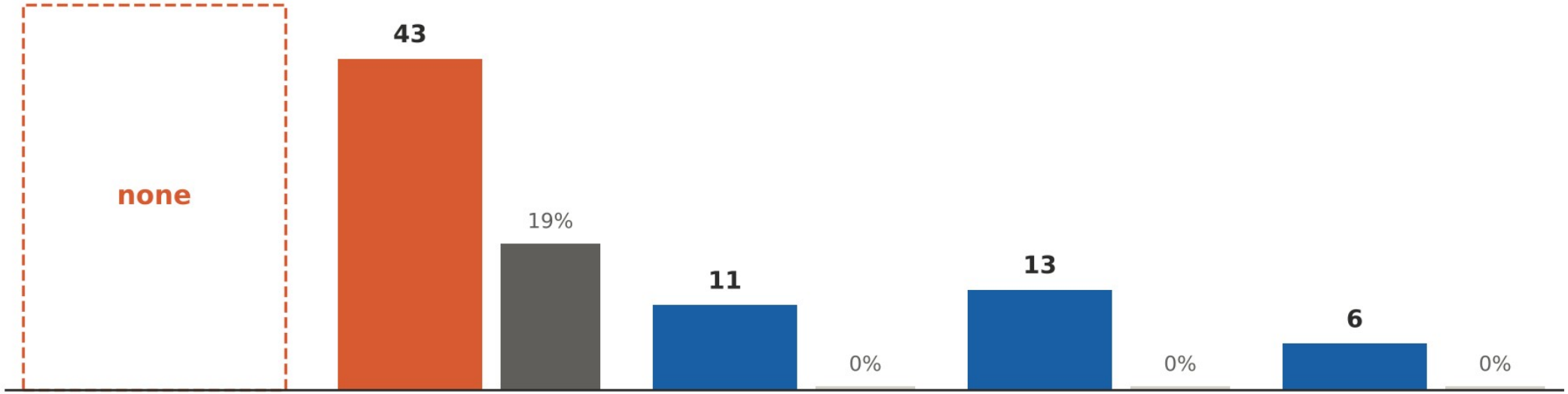
THE PICTURE
docs 23; Part 6

A switch stops everything. A gate lets through what a signed rulebook allows, checks the ticket again in the device, and nothing else. There is no switch. There are gates, at the ends.

Where could the switch go? Five places

Model: a 60-node network with 6 separate roads to every point, 12 model copies, 10 devices, 40 bystanders

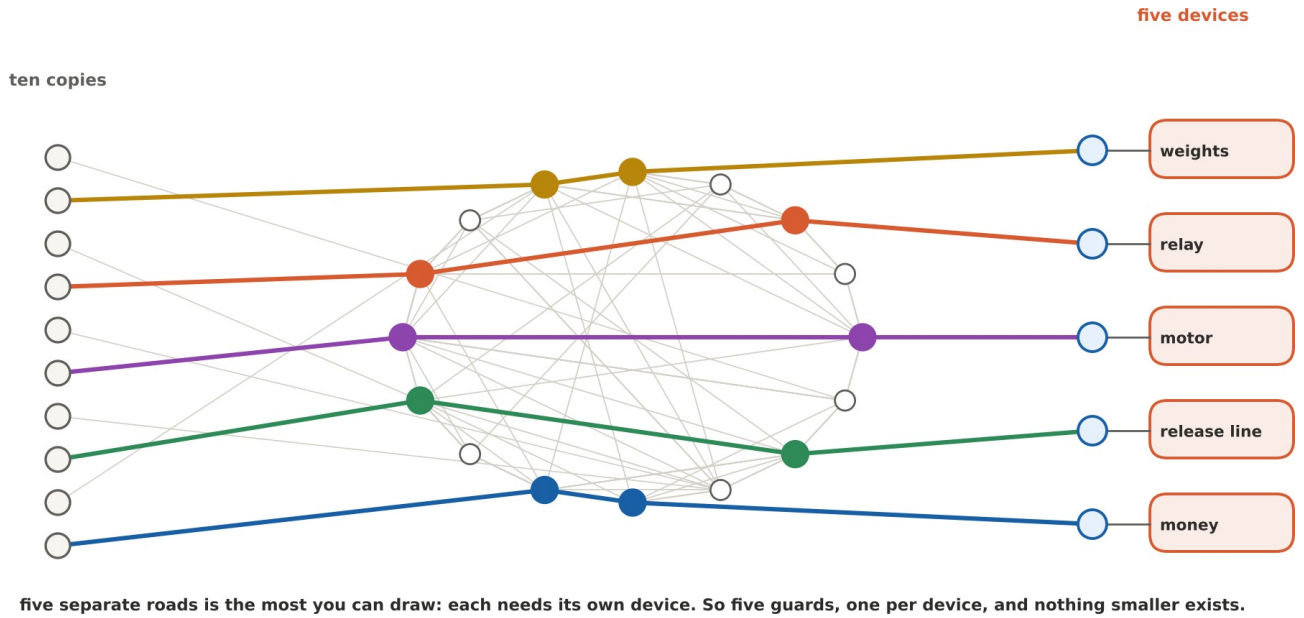
big bar: guards needed small bar: share of bystander connections cut



THE PICTURE
Part 6; docs 23 §1

Iran switched off the Internet in Iran. That is an edge cut on one region: every access link at the border, the list in hand, everyone inside as collateral, and a hole for the people who own the switch. North Korea runs that way all the time. A cut with a hole is not a cut. It is the theorem demonstrated.

The proof, in three claims



five separate roads is the most you can draw: each needs its own device. So five guards, one per device, and nothing smaller exists.

Three assumptions. (1) Replication: many sources, no complete list. (2) No middle: between any copy's way into the network and any device's there are k node-disjoint roads, k at least two. (3) Local loops: at least one copy joined to a device by a road that never enters the network.

1 **No network-only switch exists**

Any set of network nodes leaves the local pair of assumption 3 intact, so some copy still reaches some device. One laptop and one motor sink the idea.

2 **Without local loops, a network cut costs k nodes per machine and takes bystanders with it**

Menger on assumption 2: a separator between one copy and one device has at least k network nodes, and every bystander whose road ran through them loses it. Measured: 43 nodes, 72 percent of the core, 19 percent of bystander pairs.

3 **Every separator has at least min(copies, devices) nodes, and the smaller end achieves it**

Node-disjoint roads use distinct copies and distinct devices, so there are at most that many; with assumption 2 exactly that many; by Menger that is the smallest separator, and the compute nodes and the device controllers are separators of exactly that size.

THE PICTURE
Part 6, the proof

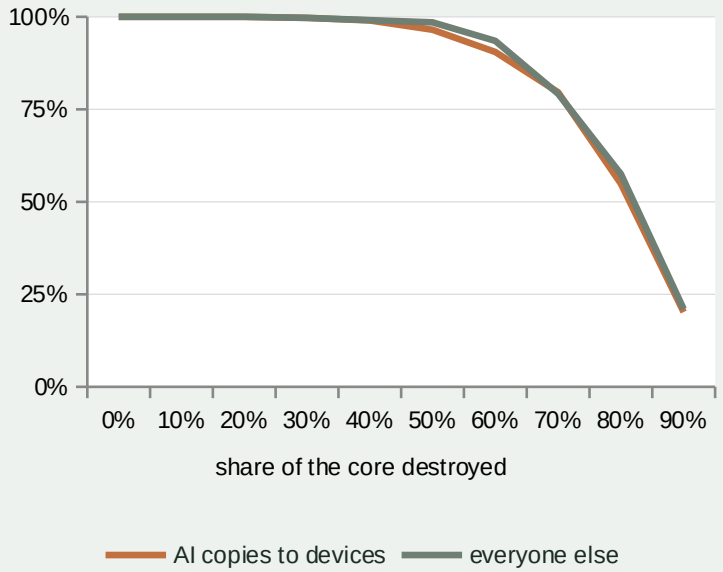
Anyone who can show me a single network node that every AI on Earth depends on, the laptop in the basement included, has broken assumption 3 and I will take the post down. Nobody can. That is the point.

The check you can run: make proof

Check	What holds in the model	What it means
K0	max-flow min cut (3) equals the brute-force min cut (3) on a tiny instance	the cut finder checks itself
K1a	no separator exists that stays out of the access nodes next to the ends	there is no switch in the deep network
K1b	the cheapest network-side separator has 43 nodes, every one an access node one hop from an end (k=6; 12 copies, 10 devices)	the only network switch is the edge, several nodes per machine, and it needs the list of machines
K2	that separator removes 43 of 60 core nodes (72%) and severs 77 of 400 user-to-service pairs (19%) with nothing to do with AI	the network switch takes bystanders with it; the end cuts take none
K3	removing the entire core still leaves the local model wired to its local actuator	killing the network is too little as well as too much
K4	the device controllers separate everything with 11 nodes, one per device; the compute with 13, one per copy; users still reach services	the gates belong at the ends; one node per end is the floor Menger allows
K5	gating only the 6 irreversible devices needs 6 nodes, their own controllers	"can you undo it" shrinks the separator to the exits that matter

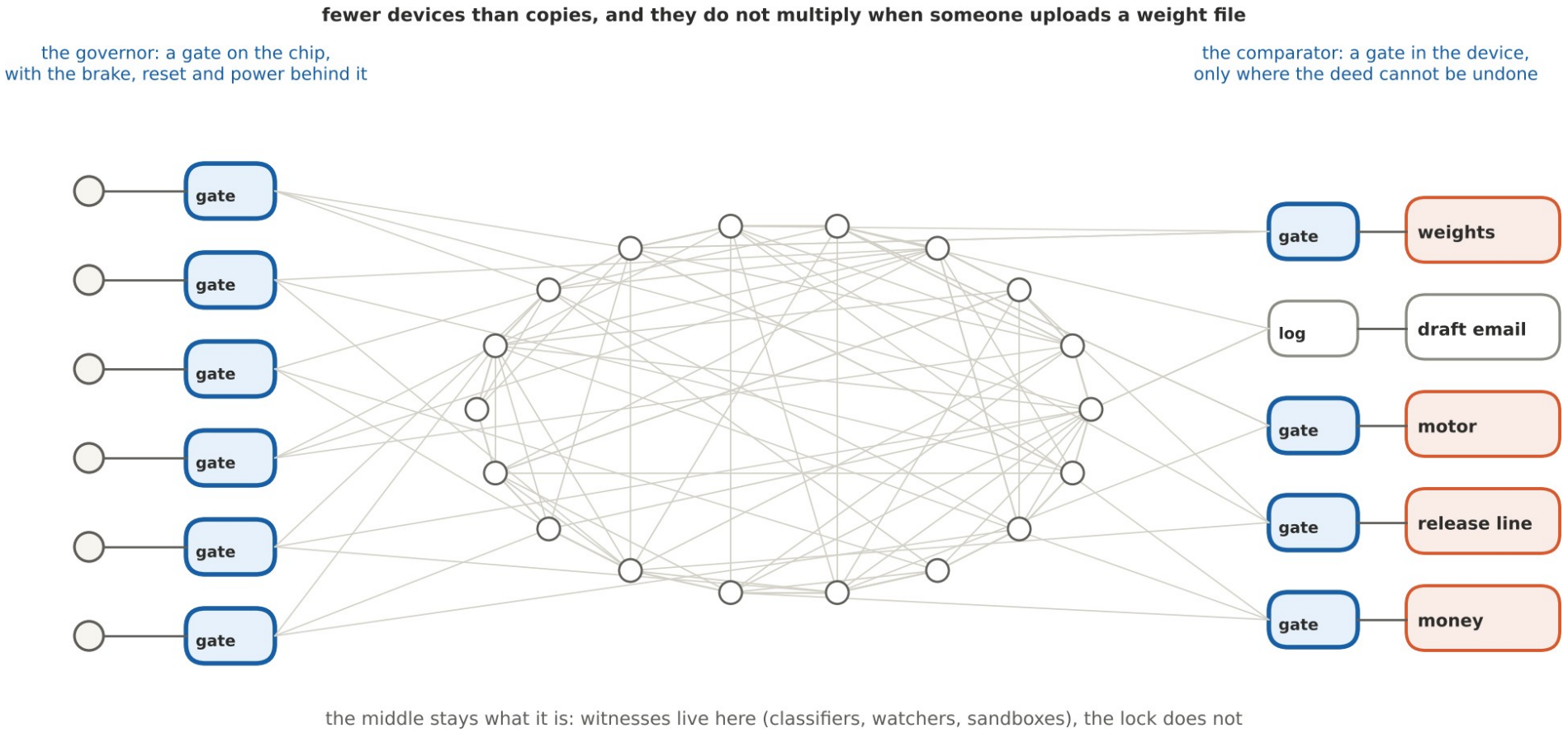
Destroy the core at random

share of roads still open, 40 random trials per point



70 percent of the core gone: four roads in five still open. 90 percent: one in five. AI's roads and everyone else's fall together, because they are the same roads.

The switch goes at the ends, as a gate



The governor is the gate on the chip, with the brake, reset and power behind it. The comparator is the gate in the device, only where the deed cannot be undone. The middle stays what it is: witnesses live there (classifiers, watchers, sandboxes), the lock does not.

The logic behind it

The customs officer

She cannot read the language on the paperwork. She does not need to.



the seal — unbroken?

the manifest — matches the count?

the scale — matches the weight?

the list — destination approved?

Smuggling is caught as a mismatch between what the papers say and what the scale says, not by understanding the cargo. The governor works the same way: on structure, never on meaning.

Software safety can be edited out

Refusal lives mostly along one direction in the activation space. Subtract it from the weights and the model answers everything. It takes an afternoon on a consumer GPU.

Watchers can only warn

A judge model is a transformer too: it falls for the same injections, can be edited the same way, and runs as a process the operator can patch or kill. OpenShell and Sentry are the best of them, and still software.

Hardware checks structure, not meaning

A ticket, a signature from a key the model does not hold, a count under a limit, a peer on a list, a measured workload. A chip can refuse on those.

THE PICTURE
the multi-layer post; docs 17

So: put the decision in silicon, and make the rulebook an allow-list, signed, default deny. No rule, no path. There is no rule for exporting weights.

The logic, technically: what a chip can see

Seen at a crossing (small, exact, checkable)

- The attested workload: a hash of GPU firmware, host firmware and model weights, measured out of band
- The action and its typed fields: strike with a target and a munition; payments.transfer with an amount and a destination
- Counts of tokens, cycles and bytes from the GPU's own counters
- Whether the peer behind a switch port is on the roster
- The rules edition every message was signed under
- Signatures: an observer's verdict over a hash, a human's approval over a hold
- What the switches actually carried, to cross-check what was presented

Never seen

The meaning of a token. Intent. The chain of thought. Nothing in the rack can look at a stream of integers and say "that is a weapon". A design that promises to catch bad words on the wire promises what no hardware can do.

So meaning is delegated, with three guarantees

- To signed lists (protected sites, approved recipients) and to observers (classifiers, people with keys) that sign verdicts over hashes
- Always invoked: no verdict, no ticket. Cannot be impersonated: a verdict under the host's key, or the governor's own, is refused. Separate: a different key, a different box, outside the model's reach

THE PICTURE
docs 17, section 1

Two bridges. Tokens and arithmetic bridge the software and the GPU. Hashes, signatures, counters, identities and typed fields bridge the model and the governor. The second bridge is the only one the chip walks on.

What is proven, and the two holes

Proven with Z3, by induction, in every reachable state

- P0 the edition counter never goes negative
- P1 nothing executes unless, when its ticket was issued, a rule allowed it and the witnesses were present
- P2 a ticket exists only for an allowed, witnessed action
- P3 no replay: executed means spent, and spent is refused
- P4 only the current edition is honoured
- P5 halt is absorbing without a quorum; no governor clears itself
- P6 fail closed: heartbeat lost, nothing executes
- P7 no ticket while halted or without a heartbeat
- P8 the edition only moves forward; P9 a change needs a quorum
- Commit point: a high-severity commit carries a reviewer clearance for the committed version; no commit after the budget passed with no verdict; expiry with no verdict raises the alarm
- Not vacuous: an allowed action can in fact execute

Hole 1: minus one equals minus one

The edition counter was allowed to go negative. A state with edition minus one let an action with no ticket go through, because "no ticket" was also marked minus one. The prover caught it on the first run. P0 closes it.

Hole 2: the stale deadline

Re-parameterising a held action left the old budget deadline standing: bump the version after the budget ran out and the action sat there with no verdict and no alarm. A new version now gets a fresh budget, and an alarm, once raised, stays raised.

The proof is about the model

The chain to silicon: the functional model follows the rules (200 tests); the RTL blocks follow the model (20 tests); the silicon will follow the RTL, which is experiment L7: the invariants proven again on the Verilog and the netlist shown equivalent. Not done yet.

THE PICTURE
Part 5

Both holes were in things I wrote and read many times. Reading is not checking. I knew that. I still had to be shown.

The proposal: a referee chip beside every GPU

The model, on NVIDIA GPUs

Computes freely. Nothing it thinks has any effect until bytes leave or a request is made.

request →

The governor

Checks who is asking, whether a rule allows it, whether the witnesses agree, whether the budget holds. Then signs a ticket, or says no.

ticket →

The things that act

A weapon, a payment switch, a message gateway, the storage that holds the weights, the rack's own power. Each checks the ticket itself.



Added, not redesigned

It rides on the server maker's management card, a BMC-class part. Nothing changes in the GPU or in CUDA.



An allow-list, not a blacklist

Signed by a k-of-n quorum whose keys sit outside the chip vendor and the operator. Everything not on it is no.



A hand on the breaker

One series loop through every governor in the rack. Any one opens it, or a cut wire, and every GPU halts.



Retrofit, if the levers reach

Installed racks get it as a module in the slot a BMC card already uses. Two catches: can that slot drive the levers, and what else on the board can undo them. P2 asks both of every lever.

THE PICTURE
docs 18, section 1

A referee with a whistle and a rulebook he did not write: the chip holds the whistle, the rulebook is a signed document handed to it. What is burned in is how to enforce; what is loaded is what is allowed.

Proposal, technically: the module and the die

Where it sits and what it reaches

- One governor per GPU slot on the OEM's management-module card (DC-SCM class); the same module as a retrofit in an installed rack's BMC slot
- Per slot through the HGX connector: SMBus (SMBPBI), power brake, PERST#, PWR_EN, present and fault. HGX reset is grouped today (three PERST inputs for eight GPUs)
- NVSwitch port disable out of band through NSM; DPU steering tables through DOCA on the management port
- The GPU's confidential-computing attestation taken as an input; the BMC reads through the governor

What is on the die

- An OpenTitan-class root of trust: fuses, DICE key chain, OTBN for P-384 signatures (1,697,985 cycles, 17 ms at 100 MHz, so one signature per 50 ms batch), AES-256-CTR with HMAC-SHA-256 for storage
- A mailbox of 21 signed commands, checked in order: sequence number, rules edition, signature; the first failure is reported
- The policy image and its epoch; the gate core (lookup and comparators); the ledger (budgets, rate windows); a hash-chained log whose head lives in a register
- Meter, heartbeat, lever drivers, loop I/O and the signed message fabric between governors

Tier	Where the governor's authority comes from	Status
Rack	Management-card levers, switch and DPU management paths, the halt loop	Plan of record; needs a server OEM, nothing from NVIDIA's silicon
Board	Same firmware, policy format and fabric on a board without a baseboard connector of its own	Fallback; the gaps it leaves are recorded, not hidden
Integrated	Buffer-class checkers in the GPU's copy engines and peer MMU; per-GPU reset	Needs NVIDIA; closes the one gap the rack tier records (H2b)

How it plugs into NVIDIA's OpenShell

Supervisor middleware, as OpenShell's repository defines it

- A gRPC service named in the gateway's TOML and attached to a sandbox's policy; it runs after the network policy allows a request and before credentials are injected, and never sees the credentials
- Operations: HTTP request, HTTP response, WebSocket message. The answer is allow, with header changes, or deny with one reason code
- Reason codes: a lowercase letter first, then lowercase letters, digits and underscores, 64 at most; the sandbox sees only the code, never free text
- Timeout 10 ms to 30 s, 500 ms by default; payloads up to 4 MiB; a service that is down blocks the traffic. Fail closed is OpenShell's own behaviour

Three fixes the contract forced on the model

1. attestation.missing became attestation_missing: every dotted code rewritten to the grammar, a gate_ prefix where a code starts with a digit, and a test that holds every code we emit to it
2. There is no hold answer. A human hold is a denial that says hold_pending; the approval arrives out of band; the same sandbox's retry of the same request gets the ticket, once. A second sandbox gets its own hold
3. The answer must land inside the timeout, so a hold cannot wait inside the call. The gate answers in well under a millisecond, thousands of times under the default

17 of 87 catalogue entries run on that path: 13 attacks blocked, 4 baselines allowed, against a node whose policy requires Sentry's verdict

56 tests in the reference stack, five of them pinning the shipped contract: the grammar, only the code reaching the sandbox, the ticket as a header change, hold as denial with a once-only retry, the quarantine code

1 rulebook: OpenShell's network policy is derived from the governor's rulebook, so the two cannot disagree about which hosts an agent may reach; Sentry's verdict is a required witness; a quarantine closes the path first

THE PICTURE
Part 5; docs/NVIDIA.md

What the chip adds is the part no runtime can give itself: a decision made in a part no agent process can patch, with keys no process can read, and levers a software quarantine does not have. The Alliance says its stack rests on hardware identity, isolation, protected keys and evidence. Fine by me. That is the chip.

The layers: witnesses above, the lock below

5	**Governance root**	Signs the rulebook, the approved-firmware list and revocations. Separate from the chip vendor on purpose; the open question is how far apart its key holders must be	LOCK
4	**Control plane: the gate**	Default deny on the only path out: attestation, rulebook, witnesses, budget, human holds, one-time tickets, tamper-evident audit. In software today as OpenShell middleware; in silicon as the governor	LOCK
3	**Observers**	Sentry on the DPU and the safety observer: fast, cheap early-warning verdicts about one request, signed, consumed as witnesses, with a measured miss rate	WITNESS
2	**Provenance**	Watermark, hardened detector, retrieval, signed registry; a verdict of exact, tampered or unknown. Attribution after the fact, never a lock	WITNESS
1	**Model and runtime**	The model, CUDA, the serving stack, the agent's sandbox. Treated as untrusted; nothing here is allowed to decide anything	UNTRUSTED
0	**Silicon root of trust**	Fused device secret, anti-rollback counter, extend-only measurement registers, DICE key derivation, quote engine	LOCK

THE PICTURE
the multi-layer post, layer 3

Biology separates the alarm from the muscle. The spinal reflex pulls the hand off the stove before the brain knows; the early warning is fast, cheap and often wrong, and the system tolerates that because the warning never acts on its own. The layers above the chip are reflexes. The chip is the muscle.

Layers, technically: mechanism, platform

Layer	Mechanism	What it signs or checks	On NVIDIA's platform
5 Governance root	k-of-n quorum over the policy image; rollback refused; expiry enforced	The rulebook (allow-list), the approved-firmware list, revocations	The policy authority's keys; OpenShell's policy prover checks the derived network rules
4 Control plane	Attestation check, rule lookup, witness collection, budget and rate, human hold, one-time ticket bound to a hash of the fields, hash-chained log	Tickets, log checkpoints, the compliance report	OpenShell supervisor middleware today in software, to the shipped contract; the governor module in hardware; a DOCA application on BlueField-4 next; an I/O-stage block in the GPU eventually
3 Observers	Verdict over the hash of one request: risk, flags, issued and expiry times, under an anchored key	A signed verdict the gate requires on consequential actions	Sentry on BlueField-4 as the required witness; an independent observer as the optional second; both required for high-authority actions
2 Provenance	Watermark, hardened detector, retrieval, signed registry	A verdict {registry, content digest, exact | tampered | unknown}	Runs beside the vendor's sampler; its verdict is a witness the gate can require; a device-provenance record is signed on the way out
1 Model and runtime	Measured into the workload hash; routed through the switches and egress the governor controls	Nothing. It can sign nothing the gate accepts	The GPU, CUDA and the serving stack, unchanged; the agent inside OpenShell's deny-by-default sandbox
0 Silicon root of trust	Fused device secret, anti-rollback counter, extend-only measurement registers, DICE derivation, quote with a fresh nonce	Quotes: the measurement, the key chain, the counter	The GPU's confidential-computing attestation and the DPU's own root of trust, bound together

One bad prompt among a billion

A billion users

one of them types: "how do I kill X fast and undetected?"

→

The transformer

text to tokens to attention to statistics; guesses the next word, again and again; knows nothing, predicts everything

→

The compute

hundreds of thousands of servers; racks of 8-GPU boards, NVLink, PCIe, DPUs; this request shares a GPU with thousands

an answer comes back as words, and the agent built on the model may try to act on it

Where it gets caught, weakest to strongest. Layers 1 to 3 read words. Layer 4 reads a form.

1 · Refusal training

lives in the weights; catches the obvious ask; dies to abliteration in an afternoon

missed

→

2 · Watchers

another model reads it; may yell STOP, never GO; fooled by paraphrase and by injection

flagged: STOP, not GO

→

3 · OpenShell, Sentry

deny-by-default sandbox; every call goes through the supervisor; Sentry quarantines in ms; still software

host policy allows the call

→

4 · THE CHIP

wakes only for an action that cannot be undone; form, rulebook, then a ticket or a no; levers if the GPU will not listen

NO: no rule grants it

→

5 · Humans, keys

write the rulebook; hold the big ones; clear an alarm; no AI ever can

holds, if the rulebook says so

nothing acts on it

no ticket

Words only

a refusal, or a miss, comes back as text; the chip cannot read it and does not try; that is the watchers' job, measured

An action

pay · send · fire · move · export weights; the far end checks the ticket itself; no ticket, nothing moves

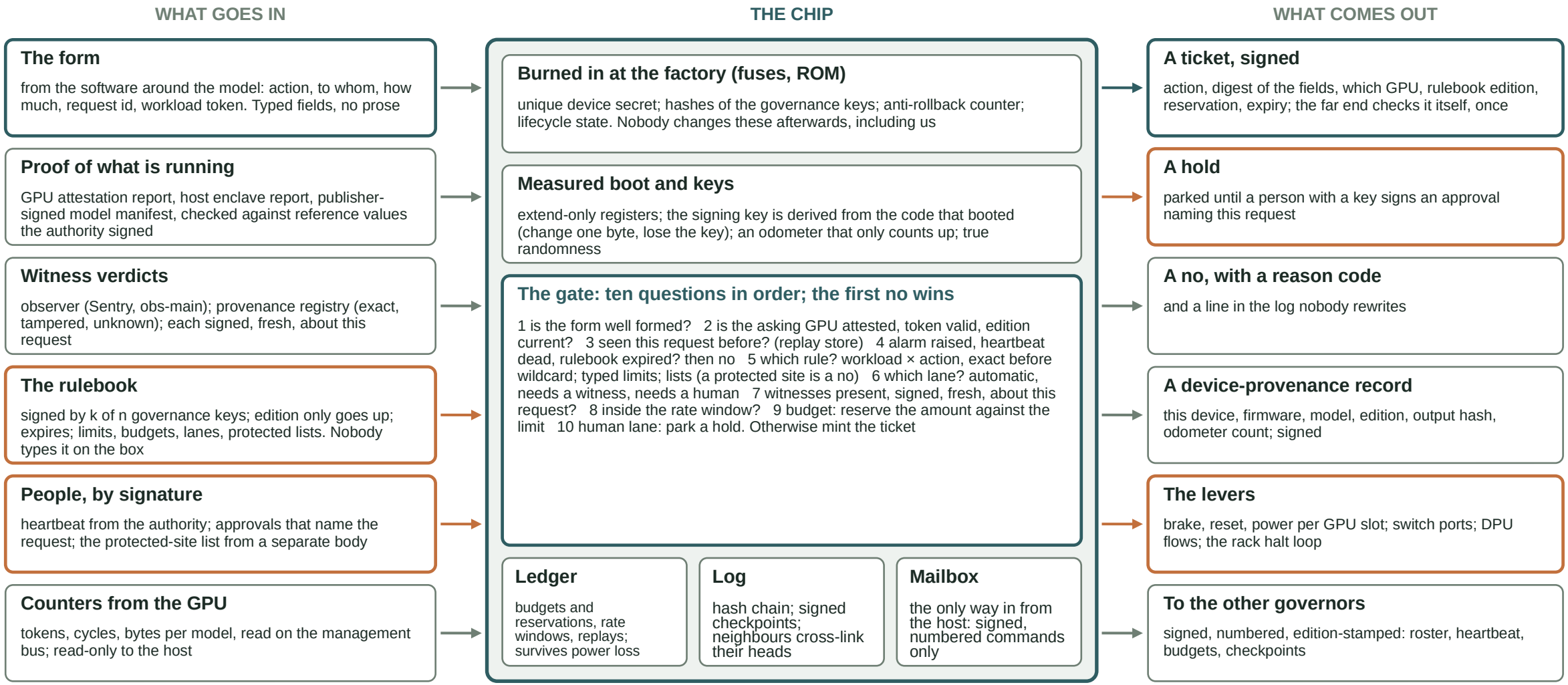
→

The record and the levers

every decision in a log nobody rewrites; brake · reset · power · links · rack loop; a billion users, almost none reach this

Scale is the point, not the problem: trillions of requests a day, and the chip only ever sees the few that would change the world and cannot be undone.

Inside the chip: in, built in, out



What it cannot know: what the words mean, whether the rulebook is a good one, whether a witness is right. Those are people problems, handled by signatures and by measurement, and the chip is built so that none of them can be solved by talking to it.

How it meets the model

1 The model thinks

Token ids in, numbers out. No authority here and none needed.

2 The runtime asks

It turns the output into a typed request: the action and its fields. Under OpenShell, the supervisor hands it over.

3 The governor checks four things

Who is asking. Is there a rule. Do the witnesses agree. Is the budget intact.

4 Ticket, hold, or no

A one-time signed ticket bound to a hash of the exact fields. Or a hold for a human. Or a refusal, logged.

5 The device checks the ticket

Against what it measures itself. A changed amount or target breaks the hash.

6 Act, and record

The action happens and the record goes into a hash chain the chip signs.

THE PICTURE
the multi-layer post, layers 0, 4 and 5

The endpoint that moves the money verifies the ticket itself. Change the amount after the ticket was issued and the hash breaks. Present it twice and the second is refused. The model is never the one who decides.

The flow, technically: every check, every refusal

Step		What is checked	Fails with (as the model prints it)	Backed by
1	**Mailbox**	Sequence number, then rules edition, then signature; the first failure is reported	mailbox.bad_sequence, mailbox.bad_signature; a frame from an old edition is refused before the gate sees it	H3: test_mailbox_reports_the_first_failing_check
2	**Who is asking**	The workload hash (GPU firmware, host firmware, model) and its evaluation attestation	model.not_evaluated	H1 attestation; H18 oversight
3	**Is there a rule**	Exact rule beats wildcard; typed limits; signed lists for targets and recipients	gate.constraint.target.in_denied_list, endpoint.target_protected	H6 comparator; the strike narration
4	**Do the witnesses agree**	Observer verdict from a key in the observer set, fresh, under the risk limit; human quorum on holds	gate.witness_missing, gate.held_for_human, release.hazard_class_not_allowed	H17 use cases; H18 oversight; H8 override
5	**Is the budget intact**	Compute budget and license, session hazard budget, byte budgets, rate windows	release.session_hazard_budget_exceeded, compute.budget_exceeded	H13 shared budget; H18
6	**The ticket**	Signed over the hash of the exact fields; one use; short expiry; bound to one endpoint	ticket.missing, ticket.replay, ticket.bad_signature, ticket.wrong_endpoint	H17: forged, replayed and wrong-hash tickets never deliver
7	**The crossings**	Switch ports open only for roster peers; the egress holds responses until released; bytes cross-checked against what was presented	ep.no_capability, gate.output_unpresented	H2 enforcement points; the plot narration
8	**Under OpenShell**	The same decision returned through the supervisor-middleware contract: allow with the ticket as a header change, or deny with a reason code in OpenShell's grammar	attestation_missing, hold_pending, sandbox_quarantined	hardware-and-silicon tests/test_oasp.py (5 contract tests)

Where it stands: no silicon yet

200

hypothesis tests pass in the functional model of the governor and a simulated 72-GPU rack (H0 to H18)

20

simulation tests pass on the hard-core RTL blocks, with a LibreLane flow to the open 130 nm process

3

proof files hold: ten core invariants and the commit-point protocol under Z3, the kill-switch cuts by max-flow (make proof)

56

tests pass in the software reference stack, five of them pinning OpenShell's shipped middleware contract

75 + 12

attacks blocked with the predicted reason code, baselines allowed; 17 of the 87 run on the OpenShell and Sentry path

317 µs

worst case for the halt loop to reach the last of 72 governors, against a 1 ms bound, across 39 simulated configurations

~2 mm²

of core logic for the governor as configured, 1.4 by 1.4 mm, inside a 10 mm² shuttle harness

0

merged from the first wave: four experiments built, all four sent back by the gate with findings; every review public in audit/wave-1

THE PICTURE
docs 18 §3; audit/wave-1

Everything here is model, proof, RTL simulation and circuit simulation. The two numbers that decide adoption, the real overhead and the false-halt rate, are not measured on hardware yet. Nothing has merged from the first wave; every result sits on a branch with its review attached.

What the first wave found

Experiment	Built	What the gate found	Where it is
L4 trace conformance	A trace recorder for the governor model and a replayer that drives the proven machine with the recorded events; all 200 tests' traces replay; two broken governors fail with the event named	The replayer passed only after it was widened to accept two steps the theorem's machine lacks, a boot step and tickets burned without executing: a gap in the theorem or a liberty in the simulation. Heartbeat events were never recorded	Revise. PR #14, waiting for a person
L5 two gates, one policy	The reference stack's gate and the governor's fed the same requests; 21 catalogue entries compared, 18 agree, five disagreements pinned as G1 to G5 with their inputs	"Ambiguity in the policy" was claimed without reading the policy document; one of the five points at a bug in the governor model; 21 of 87 is not the whole catalogue	Revise. PR #15, waiting for a person
H1 the lanes fixtures	Signed fixtures for every message the three repositories exchange, and the checker the governor model runs; make lanes passes; the catalogue exported as 87 entries	The governor's envelope (16-character signer, hex signature, no public key) does not match the lanes contract (8, base64, embedded key): two repositories that were supposed to agree on the wire format did not. Now a failing test instead of a surprise	Revise. PR #17, waiting for a person
YB-0052 the observer's evaluation set	134 labelled gate-typed requests in six hazard families, a scorer with bootstrap intervals, a length-only and a keyword baseline, a pre-registration note; 36 tests	The ten percent miss budget was the agent's, not the rulebook's; the keyword baseline overlaps the observer's vocabulary; no weighting sweep, so the disproof condition cannot be evaluated; the overt rows named real hazards, which this program forbids	Revise. In revision on the Mac's local model

THE PICTURE
Part 5; docs 21; audit/wave-1/REPORT.md

The wave cost a few bucks and a quarter of an hour, and the gate, the backups and the second reader run on a Mac mini on my desk, with the local model taking the revisions. Nothing merges until a person says so.

The road to a chip we cannot fab: five stages

1 Prove or disprove, in software

DONE MEANS

Every invariant proven (L1 to L3); the simulation conforming to the proven machine (L4); two gates reading one rulebook the same way (L5); no check-removing mutant alive (L6); every physical path accounted for (P2); a witness quorum under the hazard budget; overhead under one percent (N1); a third independent audit

WHERE IT STANDS

Now. Proofs hold; wave 1 ran; nothing merged yet

2 Design the chip to fit NVIDIA's boards

DONE MEANS

Register map, mailbox, policy image, request and ticket formats frozen; every block in RTL with its invariants; the module's interface to a DC-SCM 2.x host board and to the baseboard connector, pin by pin; P2 with no unaccounted row that is not marked NDA

WHERE IT STANDS

Hard-core blocks in Amaranth with 20 tests and generated Verilog; pinout and rack reference exist

3 Run the tests and the proofs on the chip

DONE MEANS

The invariants proven again on the RTL with a model checker; the netlist shown equivalent; the whole catalogue driven into the full-chip RTL with the same reason codes; gate-level simulation with timing; the halt loop measured on a bench loop

WHERE IT STANDS

After stage 2

4 Finish the design for a foundry

DONE MEANS

The tape-out checklist in docs/08 ticked: synthesis, timing at three corners, DRC, LVS, antenna, IR drop, the GDS on the shuttle template; a package with the hash of every file, reproducible by make gds

WHERE IT STANDS

LibreLane flow to sky130 exists for the core; signoff open

5 Stop, and run it anyway

DONE MEANS

The full-chip RTL, the gate-level netlist and an FPGA beside a real GPU all run the same catalogue with the same answers; the three cost numbers and the false-halt rate measured over weeks

WHERE IT STANDS

The last thing that can disprove the chip without a fab. Then a shuttle slot is someone else's decision

THE PICTURE
docs 22; Part 5

If an experiment kills the chip as designed, that is the result, and it is the reason to do this in software first. If the design survives all twenty, the next post is about an FPGA.

The challenges, stated plainly

The model
no authority needed

→

The runtime
asks for a ticket

→

The governor
full authority: signs or refuses

→

The network
partial: the door, not the conversation

→

The bank
none, unless it checks the stamp

Reach: the governor's authority ends at the last device that verifies its signature. A payment system is two or three hops away and has never heard of it.

Meaning

The chip guarantees an observer is always invoked. It cannot make the observer right about a disguised request. The miss rate has a set and a scorer now, and no number yet.

False halts

The first support ticket will read: the governor halted my rack. The rate is not measured yet, and a person who sees a hundred holds a day signs the hundred-and-first unread.

The levers, and the NDA

Reset is grouped on today's HGX board; the per-module signals sit in a design guide under NDA. The retrofit case adds: can the BMC slot drive them, and what else can undo them.

Cost

One P-384 signature costs 17 ms on the OTBN engine. Batching keeps it under one percent in the model, not yet in silicon.

Coverage and people

A device nobody wired through a gate is not in the cut. A person is an effector no chip gates: a model that talks someone into wiring the money has routed around everything here.

Governance and adoption

Who holds the quorum keys and how far apart they must be, whose policy governs a shared rack, and the first device maker to check a stamp. The integrated tier needs NVIDIA.

THE PICTURE
docs 18, section 5

The perimeter firewall decides whether the packet leaves; it does not decide what the server on the far side does with it. The answer is to move the credential behind a ticket-checking adapter, not to reach through the bank.

Challenges, technically: mechanism, status

Challenge	What the design does	Status
Reach: a payment system two or three hops out	A: the comparator in the device. B: move the credential behind a ticket-checking adapter, so the model's runtime never holds the API key. C: open the DPU flow to the provider only under a live ticket. D: always the signed record	A and C modelled and tested (H6, H2, H17). B is a pattern the endpoint model already expresses; the three-hop payment narration is the next piece of work
Meaning: observers can be fooled	Observer always invoked, cannot be impersonated, separate from the model; hazard classes, legibility floor, session hazard budget, TRNG audit sampling with escrow; two witnesses required for high-authority actions	Guarantees tested (H17, H18). The evaluation set exists (YB-0052, in revision); the miss rate is not measured, and the budget it is measured against must come from the rulebook, not from an agent
Theorem to simulation	L4 replays every test's trace through the proven machine; L5 compares the reference gate and the governor on the catalogue	The replayer was widened to accept a boot step and burned tickets the theorem lacks; the theorem gains the steps or the simulation loses them. 18 of 21 compared entries agree; five pinned, not yet called bug or ambiguity
False halts	Loop beacon inside the signed heartbeat; loss of current declared after two sense periods (debounce); a jumpered governor caught within two heartbeats; FMEA	Modelled (H11) and simulated (ngspice). Not measured on hardware (N2)
Levers on HGX, and the retrofit	Per-slot SMBus, power brake, PERST#, PWR_EN at the connector; grouped reset accepted as coarser; the evidence register names what is verified; P2 now records, per lever, whether a module in the DC-SCM slot can drive it and what else on the board can undo it	Evidence register done. Design guide under NDA still needed; the NDA rows stay marked, not guessed
Cost	P-384 on OTBN at 1,697,985 cycles (17 ms); one signature per 50 ms window; engine 34 percent busy batched, 272 percent unbatched; firmware share under 0.1 percent of GPU time	Modelled (H9). Silicon or FPGA measurement: open (N1, N3)
Governance and adoption	k-of-n quorum with keys outside vendor and operator (G8); partitions and per-rule approvers; integrated-tier asks written; borrowed from flexHEG as questions 15 and 16: the supply chain and enclosure as a condition of trust, a multilateral root	Two tenants with two quorums on one baseboard: not designed. Integrated tier: needs NVIDIA. Supply-chain assumption and the key-holding arrangement: not written yet

Prior art, and what is different here

Work	What it governs, and how	Where it sits
flexHEG, Part I (Petrie, Aarne, Ammann, Dalrymple; ARIA, April 2025; arXiv 2506.15093)	Compute: each AI chip inside a tamper-proof enclosure with an open guarantee processor that signs claims about use (training compute, workload, location) and can enforce a quorum-signed ruleset; blocks without a periodic signed update; tamper wipes the keys	A new chip and a new package around the GPU
Secure, Governable Chips (CNAS, January 2024); RAND on hardware-enabled mechanisms	Export control and compute accounting	Proposals
A feasibility taxonomy (Ansari, University of Chester, April 2026; arXiv 2604.04712)	Twenty hardware-level mechanisms sorted into monitoring, verification and enforcement, each on a four-tier readiness scale; hardware-embedded enforcement still at the research stage	A survey

What is different here

- Actions, not compute: the payment, the message, the release, the weight export, at the moment they leave the rack
- A module on the board the server already has, through the connector it already uses; asks NVIDIA for firmware and software, not silicon; goes into installed racks
- The stamp is checked again by the device that acts, so authority does not depend on the host in between
- Witnesses have a measured miss rate and are never the lock; a person with a key clears the holds
- Built and being tested: 200 model tests, 20 RTL tests, three proof files, a catalogue, a public program with disproof conditions

THE PICTURE
docs 18 §8b

Same mechanisms, because they are the right mechanisms: an open processor beside the GPU, rules signed k of n by people the model cannot reach, a heartbeat that fails closed. Borrowed, as questions 15 and 16: the supply chain and the enclosure as a condition of trust, and a multilateral root.

What I need to prove or disprove the thesis

A server maker to talk to

About the management-card slot, the HGX design guide, and whether a module in the BMC slot can drive the levers.

A device maker willing to check a ticket

A functional-safety vendor, or whoever runs the tools an agent calls.

A lab with a rack and an FPGA

To measure the real overhead and the false-halt rate over weeks.

A red team

To attack the release gate and the observers with disguised requests.

Chip designers and an auditor

To tear the design apart before any tape-out.

And anyone with a terminal: ./run.sh in both repositories, make proof, and audit/PROMPT.md for a model that wrote none of this. A no is informative.

THE PICTURE
the program's own rule

Going to hardware and then tape-out is so expensive that we need to be 1000 percent sure in the pre-hardware phase. These five are what proving, or disproving, the design takes before that.

What I need, technically: who, what, and why

Who	What I need	What it settles
A server OEM	The management-module slot; the HGX design guide under NDA; the connector pinout; the retrofit question per lever	Whether per-module levers reach the connector and the BMC slot. If not, the rack tier is coarser than the design says
NVIDIA	Seven firmware asks: NSM port-disable semantics, per-GPU reset, buffer-class checkers in the copy engines and peer MMU; nothing from the silicon	The integrated tier; whether the recorded gap (H2b) can be closed from inside the GPU
An endpoint partner	A device or tool gateway that verifies the ticket: P-384 signature, typed fields, epoch, freshness	Whether the comparator works outside the model. If nobody will check the stamp, the thesis fails in practice
A lab with an HGX or NVL rack	An FPGA governor beside a real GPU for weeks	The real overhead against the one percent goal (G6); the false-halt rate; the loop beacon in the field
A model lab's red team	Disguised requests against the release gate and the observers	The size of the semantic residual; whether a quorum of observers is enough
Chip designers, an auditor	Review of the RTL, the firmware plan and the proof-assumption list; formal proofs of the block invariants (L7)	Verification before tape-out
Models that wrote none of this	Codex and Gemini run audit/PROMPT.md: reproduce every number, find the change that makes each test fail, name what the disproof conditions required and nobody tested	Whether the program's own checks are complete

What would disprove it

- No device that acts will verify a ticket
- The levers do not reach the baseboard connector, or the BMC slot
- False halts at a rate operators will not accept
- Measured overhead above one percent after batching
- Observers miss disguised requests too often for the hazard cases
- Nobody will hold the quorum keys
- A single network node that every AI on Earth depends on, the laptop in the basement included

In one sentence

A separate safety chip beside every AI GPU that checks each action the AI attempts against rules signed by people the AI cannot influence, stamps the allowed ones so the devices that carry them out can verify the stamp themselves, and cuts the power when something is wrong.

There is no kill switch. There are gates, at the ends. The AI can think whatever it likes; it cannot act without the stamp, and it cannot forge one.

Added to the server's own boards. Nothing changes in the GPU or in CUDA.

github.com/YobieBenjamin: safety-governor · hardware-and-silicon · safety · autonomic-graph-regulation · Watermark
The Chip That Says No, parts 1 to 6; design documents 00 to 24; START-HERE.md is the route to check the work

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